Higher National Diploma in Engineering (Electrical)

Fourth Year, First Semester Examination – 2018

EE3216- Internet Technologies

Instructions for Candidates:	No. of questions : 06
Answer five(05) questions only	No. of pages : 02
all questions carry equal marks	Time :Three(03) Hours

Q 01.

i). What is an internet service? List four services of internet. **(05 Marks)**

ii). Compare and contrast the two types of web content. **(05 Marks)**

iii). Briefly describe the role of ISP in Internet connectivity. **(05 Marks)**

iv). List five RIR with the area covered. **(05 Marks)**

Q 02.

i). Draw the TCP/IP model with protocols. **(04 Marks)**

ii). Describe the following terms in TCP/IP. **(06 Marks)**

a) Port

b) Process

c) Host

iii). What is a name server? List three types of it. **(04 Marks)**

iv). Describe the use of daemon in TCP/IP by giving suitable examples **(06 Marks)**

Q 03.

i). What is cyber-attack? Give two examples. **(04 Marks)**

ii). Briefly describe the differences between symmetric and asymmetric cryptography. **(06 Marks)**

iii). List four steps that you can take as precautions to protect your data from the cyber-attack. **(04 Marks)**

iv). Describe the following with regard to email Security. **(06 Marks)**

a) Spam.
b) Phishing
c) Virus

Q 04.

i). Briefly describe the protocols used in an email system. **(06 Marks)**

ii). Describe the functions of one of the protocols that you mentioned in part (i) with a diagram. **(04 Marks)**

iii). Compare and contrast streaming and webcasting. **(04 Marks)**

iv). Briefly describe the use of the following protocols. **(06 Marks)**

a) Media Gateway Control Protocol (MGCP)
b) Real Time Streaming Protocol (RTSP)

Q 05.

The "fast Deliver courier "company want to maintain its shipping details in an xml database. The requirements are as follows. They maintain all the shipping details under ship Detail database which has order id for each shipping. Each shipping contains order person, shipping location and the item. Order person has a name. Shipping location should track the name, address, city and country. Each item has a title, quantity, price and note.

i). Write down the XML schema for the above data. **(08 Marks)**

ii). Write down the XML file with sample data that you are supposed to store. **(08 Marks)**

iii). Draw the XML tree for the above XML file. **(04 Marks)**

Q 06.

Write short notes on following topics. **(4 x 5 Marks)**

a) Domain Name System
b) MIME
c) Digital Signature
d) Socket